

Self Study

Group BY Problems

1.586 Customer Placing the Largest Number of Orders

DescriptionEditorialSolutionsSubmissions

586. Customer Placing the Largest Number of Orders

Solved

EasyTopicsCompaniesHint

[SQL Schema](#) > [Pandas Schema](#) >

Table: Orders

Column Name	Type
order_number	int
customer_number	int

order_number is the primary key (column with unique values) for this table.
This table contains information about the order ID and the customer ID.

Write a solution to find the `customer_number` for the customer who has placed the **largest number of orders**.

The test cases are generated so that **exactly one customer** will have placed more orders than any other customer.

The result format is in the following example.

CodeAccepted X

MySQLAuto

```
1 # Write your MySQL query statement below
2
3 SELECT customer_number
4 FROM Orders
5 GROUP BY customer_number
6 ORDER BY COUNT(customer_number) DESC
7 LIMIT 1;
```

SavedLn 7, Col 9

2. 619 Biggest Single Number

619. Biggest Single Number

Solved

EasyTopicsCompanies

[SQL Schema](#) > [Pandas Schema](#) >

Table: MyNumbers

Column Name	Type
num	int

This table may contain duplicates (In other words, there is no primary key for this table in SQL).
Each row of this table contains an integer.

A **single number** is a number that appeared only once in the `MyNumbers` table.

Find the largest **single number**. If there is no **single number**, report `null`.

MySQLAuto

```
1 # Write your MySQL query statement below
2
3 select MAX(num) as num
4 from MyNumbers
5 where num in(select num
6               from MyNumbers
7               group by num
8               having count(*)=1);
9
```

SavedLn 2, Col 1

3. 511. Game Play Analysis I

511. Game Play Analysis I

Solved

EasyTopicsCompanies

[SQL Schema](#) > [Pandas Schema](#) >

Table: Activity

Column Name	Type
player_id	int
device_id	int
event_date	date
games_played	int

(player_id, event_date) is the primary key (combination of columns with unique values) of this table.
This table shows the activity of players of some games.
Each row is a record of a player who logged in and played a number of games (possibly 0) before logging out on someday using some device.

MySQLAuto

```
1 # Write your MySQL query statement below
2
3 select player_id,min(event_date) as first_login
4 from Activity
5 group by player_id
```

Ln 4, Col 1

4. 1693. Daily Leads and Partners

1693. Daily Leads and Partners

Solved

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: DailySales

Column Name	Type
date_id	date
make_name	varchar
lead_id	int
partner_id	int

There is no primary key (column with unique values) for this table. It may contain duplicates.
This table contains the date and the name of the product sold and the IDs of the lead and partner it was sold to.
The name consists of only lowercase English letters.

For each `date_id` and `make_name`, find the number of **distinct** `lead_id`'s and **distinct** `partner_id`'s.
Return the result table in **any order**.
The result format is in the following example.

MySQL Auto

```
1 # Write your MySQL query statement below
2
3 SELECT
4     date_id,
5     make_name,
6     COUNT(DISTINCT lead_id) AS unique_leads,
7     COUNT(DISTINCT partner_id) AS unique_partners
8 FROM DailySales
9 GROUP BY date_id, make_name;
```

Saved Ln 9, Col 29

5. 1741. Find Total Time Spent by Each Employee

1741. Find Total Time Spent by Each Employee

Solved

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Employees

Column Name	Type
emp_id	int
event_day	date
in_time	int
out_time	int

(`emp_id`, `event_day`, `in_time`) is the primary key (combinations of columns with unique values) of this table.
The table shows the employees' entries and exits in an office.
`event_day` is the day at which this event happened, `in_time` is the minute at which the employee entered the office, and `out_time` is the minute at which they left the office.
`in_time` and `out_time` are between 1 and 1440.
It is guaranteed that no two events on the same day intersect in time, and `in_time` < `out_time`.

Write a solution to calculate the total time in **minutes** spent by each employee on each day at the office. Note that within one day, an employee can enter and leave more than once. The time spent in the office for a single entry is

MySQL Auto

```
1 # Write your MySQL query statement below
2
3 select
4     event_day as day, emp_id
5     , sum(out_time-in_time) as total_time
6 from
7     Employees
8 group by
9     event_day, emp_id
```

Saved Ln 9, Col 22

823 50 6 Online Testcase Test Result

6. 1050. Actors and Directors Who Cooperated at Least Three Times

1050. Actors and Directors Who Cooperated At Least Three Times

Solved

Easy Topics Companies

SQL Schema Pandas Schema

Table: ActorDirector

Column Name	Type
actor_id	int
director_id	int
timestamp	int

timestamp is the primary key (column with unique values) for this table.

Write a solution to find all the pairs (actor_id, director_id) where the actor has cooperated with the director at least three times.

Return the result table in **any order**.

The result format is in the following example.

Example 1:

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT actor_id,director_id
4 FROM ActorDirector
5 GROUP BY actor_id,director_id
6 HAVING COUNT(timestamp) >=3;

```

Restored from local Upgrade to Cloud Saving

Ln 1, Col 1

7. 1890. The Latest Login in 2020

1890. The Latest Login in 2020

Solved

Easy Topics Companies

SQL Schema Pandas Schema

Table: Logins

Column Name	Type
user_id	int
time_stamp	datetime

(user_id, time_stamp) is the primary key (combination of columns with unique values) for this table.

Each row contains information about the login time for the user with ID user_id.

Write a solution to report the **latest** login for all users in the year 2020. Do **not** include the users who did not login in 2020.

Return the result table in **any order**.

The result format is in the following example.

Example 1:

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 select user_id,
4 max(time_stamp) as last_stamp
5 from logins
6 where year(time_stamp)=2020
7 group by user_id;

```

Saved

Ln 7, Col 18

8. 1211. Queries Quality and Percentage

1211. Queries Quality and Percentage

Solved

Easy Topics Companies

SQL Schema Pandas Schema

Table: Queries

Column Name	Type
query_name	varchar
result	varchar
position	int
rating	int

This table may have duplicate rows.

This table contains information collected from some queries on a database.

The position column has a value from 1 to 500.

The rating column has a value from 1 to 5. Query with rating less than 3 is a poor query.

We define query **quality** as:

The average of the ratio between query rating and its position.

We also define **poor query percentage** as:

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 select
4 query_name,
5 round(avg(cast(rating as decimal) / position), 2) as quality,
6 round(sum(case when rating < 3 then 1 else 0 end) * 100 / count(*), 2) as poor_query_percentage
7 from
8 queries
9 group by
10 query_name;

```

Saved

Ln 10, Col 12

9. 1141. User Activity for the Past 30 Days

1141. User Activity for the Past 30 Days I

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Activity

Column Name	Type
user_id	int
session_id	int
activity_date	date
activity_type	enum

This table may have duplicate rows.
The activity_type column is an ENUM (category) of type ('open_session', 'end_session', 'scroll_down', 'send_message').
The table shows the user activities for a social media website.
Note that each session belongs to exactly one user.

Write a solution to find the daily active user count for a period of 30 days ending 2019-07-27 inclusively. A user was active on someday if they made at least one activity on that day.

Return the result table in **any order**.

The result format is in the following example.

```
1 # Write your MySQL query statement below
2
3
4 SELECT
5     a.activity_date AS day,
6     COUNT(DISTINCT user_id) AS active_users
7 FROM activity a
8 WHERE activity_date BETWEEN DATE_SUB('2019-07-27', INTERVAL 29 DAY) AND '2019-07-27'
9 GROUP BY activity_date;
10
```

10. 596. Classes With At Least 5 Students

596. Classes With at Least 5 Students

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Courses

Column Name	Type
student	varchar
class	varchar

(student, class) is the primary key (combination of columns with unique values) for this table.
Each row of this table indicates the name of a student and the class in which they are enrolled.

Write a solution to find all the classes that have **at least five students**.

Return the result table in **any order**.

The result format is in the following example.

Example 1:

```
1 # Write your MySQL query statement below
2
3 SELECT class
4 FROM courses
5 GROUP BY class
6 HAVING COUNT(student) >= 5;
```

Joins Problem

1. Replace Employee ID With The Unique Identifier

1378. Replace Employee ID With The Unique Identifier

Solved

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Employees

Column Name	Type
id	int
name	varchar

id is the primary key (column with unique values) for this table.
Each row of this table contains the id and the name of an employee in a company.

Table: EmployeeUNI

Column Name	Type
id	int
unique_id	int

(id, unique_id) is the primary key (combination of columns with unique values) for this table.
Each row of this table contains the id and the corresponding unique id of an

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT unique_id,name FROM Employees e LEFT OUTER JOIN EmployeeUNI u ON e.id = u.id;

```

Restored from local Upgrade to Cloud Saving

Ln 1, Col 1

2. Product Sales Analysis I

1068. Product Sales Analysis I

Solved

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Sales

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

(sale_id, year) is the primary key (combination of columns with unique values) of this table.
product_id is a foreign key (reference column) to Product table.
Each row of this table shows a sale on the product product_id in a certain year.
Note that the price is per unit.

Table: Product

Column Name	Type
-------------	------

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT p.product_name,s.year,s.price
4 FROM sales s INNER JOIN product p
5 ON s.product_id = p.product_id;

```

Restored from local Upgrade to Cloud Saving

Ln 1, Col 1

3. Employee Bonus

577. Employee Bonus

Solved

Easy Topics Companies Hint

SQL Schema > Pandas Schema >

Table: Employee

Column Name	Type
empId	int
name	varchar
supervisor	int
salary	int

empId is the column with unique values for this table.
Each row of this table indicates the name and the ID of an employee in addition to their salary and the id of their manager.

Table: Bonus

Column Name	Type
empId	int
bonus	int

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT e.name, b.bonus
4 FROM Employee e
5 LEFT JOIN Bonus b
6 ON e.empId = b.empId
7 WHERE b.bonus < 1000
8 OR b.bonus IS NULL;

```

Saved

Ln 8, Col 23

4. Students and Examinations

1280. Students and Examinations

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Students

Column Name	Type
student_id	int
student_name	varchar

student_id is the primary key (column with unique values) for this table.
Each row of this table contains the ID and the name of one student in the school.

Table: Subjects

Column Name	Type
subject_name	varchar

subject_name is the primary key (column with unique values) for this table.
Each row of this table contains the name of one subject in the school.

MySQL Auto

```
1 /* Write your T-SQL query statement below */
2
3 SELECT
4     s.student_id,
5     s.student_name,
6     sub.subject_name,
7     COUNT(e.subject_name) AS attended_exams
8 FROM Students s
9 CROSS JOIN Subjects sub
10 LEFT JOIN Examinations e
11     ON e.student_id = s.student_id
12     AND sub.subject_name = e.subject_name
13 GROUP BY
14     s.student_id,
15     s.student_name,
16     sub.subject_name
17 ORDER BY
18     s.student_id,
19     sub.subject_name;
```

Saved Ln 19, Col 22

5. Managers With At Least 5 Direct Reports

570. Managers with at Least 5 Direct Reports

Medium Topics Companies Hint

SQL Schema > Pandas Schema >

Table: Employee

Column Name	Type
id	int
name	varchar
department	varchar
managerId	int

id is the primary key (column with unique values) for this table.
Each row of this table indicates the name of an employee, their department, and the id of their manager.
If managerId is null, then the employee does not have a manager.
No employee will be the manager of themselves.

Write a solution to find managers with at least **five direct reports**.

Return the result table in **any order**.

The result format is in the following example.

MySQL Auto

```
1 # Write your MySQL query statement below
2
3 SELECT E1.name
4 FROM Employee E1
5 JOIN (
6     SELECT managerId, COUNT(*) AS directReports
7 FROM Employee
8 GROUP BY managerId
9 HAVING COUNT(*) >= 5
10 ) E2 ON E1.id = E2.managerId;
```

Saved Ln 10, Col 30

6. Asian Population

Change The

```
1 select sum(city.population)
2 from city inner join country
3 on city.countrycode = country.code
4 where continent = 'Asia';
5
```

7. Average Population of Each Continent

Change Theme Language MySQL

```
1 SELECT (COUNTRY.Continent), floor(avg(CITY.Population))
2 from COUNTRY join CITY
3 on CITY.CountryCode = COUNTRY.Code
4 group by COUNTRY.Continent
```

8. Last Person to Fit in the Bus

1204. Last Person to Fit in the Bus

Solved

Medium Topics Companies

SQL Schema > Pandas Schema >

Table: Queue

Column Name	Type
person_id	int
person_name	varchar
weight	int
turn	int

person_id column contains unique values.
This table has the information about all people waiting for a bus.
The person_id and turn columns will contain all numbers from 1 to n, where n is the number of rows in the table.
turn determines the order of which the people will board the bus, where turn=1 denotes the first person to board and turn-n denotes the last person to board.
weight is the weight of the person in kilograms.

There is a queue of people waiting to board a bus. However, the bus has a weight limit of 1000 kilograms, so there may be some people who cannot board.

MySQL Auto

```
1 # Write your MySQL query statement below
2 SELECT person_name
3 FROM (
4     SELECT person_name, turn,
5           SUM(weight) OVER (ORDER BY turn) AS total_weight
6     FROM Queue
7 ) AS q
8 WHERE total_weight <= 1000
9 ORDER BY turn DESC
10 LIMIT 1;
```

Saved In 1, Col 1

9. Product Price at a Given Date

1164. Product Price at a Given Date

Solved

Medium Topics Companies

SQL Schema Pandas Schema

Table: Products

Column Name	Type
product_id	int
new_price	int
change_date	date

(product_id, change_date) is the primary key (combination of columns with unique values) of this table.
Each row of this table indicates that the price of some product was changed to a new price at some date.

Initially, all products have price 10.

Write a solution to find the prices of all products on the date `2019-08-16`.

Return the result table in **any order**.

The result format is in the following example.

```

1 # Write your MySQL query statement below
2
3 SELECT
4     p.product_id
5     ,COALESCE(
6         SELECT pr.new_price
7         FROM products pr
8         WHERE p.product_id=pr.product_id
9               AND pr.change_date<='2019-08-16'
10        ORDER BY pr.change_date DESC LIMIT 1
11     ),10) AS price
12 FROM (
13     SELECT DISTINCT product_id
14     FROM products
15 ) AS p

```

Saved

Ln 15, Col 7

Subquery Problems

1. 181. Employees Earning More Than Their Managers

181. Employees Earning More Than Their Managers

Solved

Easy Topics Companies

SQL Schema Pandas Schema

Table: Employee

Column Name	Type
id	int
name	varchar
salary	int
managerId	int

id is the primary key (column with unique values) for this table.
Each row of this table indicates the ID of an employee, their name, salary, and the ID of their manager.

Write a solution to find the employees who earn more than their managers.

Return the result table in **any order**.

The result format is in the following example.

```

1 # Write your MySQL query statement below
2
3 SELECT e2.name as Employee
4 FROM employee e1
5 INNER JOIN employee e2 ON e1.id = e2.managerID
6 WHERE
7     e1.salary < e2.salary

```

Restored from local Upgrade to Cloud Saving

Ln 1, Col 1

2. 183. Customers Who Never Order

183. Customers Who Never Order

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Customers

Column Name	Type
id	int
name	varchar

id is the primary key (column with unique values) for this table. Each row of this table indicates the ID and name of a customer.

Table: Orders

Column Name	Type
id	int
customerId	int

id is the primary key (column with unique values) for this table. customerId is a foreign key (reference columns) of the ID from the Customers table. Each row of this table indicates the ID of an order and the ID of the customer who

MySQL Auto

```

1 # Write your MySQL query statement below
2 SELECT name AS Customers
3 FROM Customers
4 LEFT JOIN Orders
5   ON customers.id = orders.customerId
6 WHERE orders.customerId IS NULL;

```

Saved Ln 1, Col 1

3. 197. Rising Temperature

197. Rising Temperature

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Weather

Column Name	Type
id	int
recordDate	date
temperature	int

id is the column with unique values for this table. There are no different rows with the same recordDate. This table contains information about the temperature on a certain day.

Write a solution to find all dates' id with higher temperatures compared to its previous dates (yesterday).

Return the result table in **any order**.

The result format is in the following example.

Example 1:

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT W1.id
4 FROM Weather W1
5 JOIN Weather W2
6   ON W1.recordDate = DATE_ADD(W2.recordDate, INTERVAL 1 DAY)
7 WHERE W1.temperature > W2.temperature;

```

Saved Ln 1, Col 1

4. 607. Sales Person

607. Sales Person

Easy Topics Companies Hint

SQL Schema > Pandas Schema >

Table: SalesPerson

Column Name	Type
sales_id	int
name	varchar
salary	int
commission_rate	int
hire_date	date

sales_id is the primary key (column with unique values) for this table. Each row of this table indicates the name and the ID of a salesperson alongside their salary, commission rate, and hire date.

Table: Company

Column Name	Type
com_id	int
name	varchar

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 select s.name
4 from SalesPerson s
5 where s.name not in
6   (select s.name
7    from SalesPerson s
8     left join Orders o on s.sales_id = o.sales_id
9     left join Company c on o.com_id = c.com_id
10    where c.name = 'Red')

```

Saved Ln 1, Col 1

5. 180. Consecutive Numbers

180. Consecutive Numbers

Medium Topics Companies

SQL Schema > Pandas Schema >

Table: Logs

Column Name	Type
id	int
num	varchar

In SQL, id is the primary key for this table.
id is an autoincrement column starting from 1.

Find all numbers that appear at least three times consecutively.

Return the result table in **any order**.

The result format is in the following example.

Example 1:

MySQL Auto

```
1 # Write your MySQL query statement below
2
3 SELECT DISTINCT log1.num AS ConsecutiveNums
4 FROM Logs AS log1, Logs AS log2, Logs AS log3
5 WHERE
6     log1.id + 1 = log2.id AND
7     log2.id + 1 = log3.id AND
8     log1.num = log2.num AND
9     log2.num = log3.num;
10
```

6. 175. Combine Two Tables

175. Combine Two Tables

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Person

Column Name	Type
personId	int
lastName	varchar
firstName	varchar

personId is the primary key (column with unique values) for this table.
This table contains information about the ID of some persons and their first and last names.

Table: Address

Column Name	Type
addressId	int
personId	int
city	varchar
state	varchar

MySQL Auto

```
1 # Write your MySQL query statement below
2 SELECT p.firstName,p.lastName,a.city,a.state
3 FROM Person p LEFT OUTER JOIN Address a
4 ON p.personId = a.personId;
5
```

Saved Ln 5, Col 1

7. 178. Rank Scores

178. Rank Scores

Solved

Medium Topics Companies

SQL Schema > Pandas Schema >

Table: Scores

Column Name	Type
id	int
score	decimal

id is the primary key (column with unique values) for this table.
Each row of this table contains the score of a game. Score is a floating point value with two decimal places.

Write a solution to find the rank of the scores. The ranking should be calculated according to the following rules:

- The scores should be ranked from the highest to the lowest.
- If there is a tie between two scores, both should have the same ranking.
- After a tie, the next ranking number should be the next consecutive integer value. In other words, there should be no holes between ranks.

Return the result table ordered by `score` in descending order.

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT S.score, COUNT(S2.SCORE) as "rank" FROM SCORES S,
4 (SELECT DISTINCT SCORE FROM SCORES) S2
5 WHERE S.SCORE <= S2.SCORE
6 GROUP BY S.ID
7 ORDER BY S.SCORE DESC;
8

```

Restored from local Upgrade to Cloud Saving

Ln 1, Col 1

8. 570. Managers with at Least 5 Direct Reports

570. Managers with at Least 5 Direct Reports

Solved

Medium Topics Companies Hint

SQL Schema > Pandas Schema >

Table: Employee

Column Name	Type
id	int
name	varchar
department	varchar
managerId	int

id is the primary key (column with unique values) for this table.
Each row of this table indicates the name of an employee, their department, and the id of their manager.
If managerId is null, then the employee does not have a manager.
No employee will be the manager of himself.

Write a solution to find managers with at least **five direct reports**.

Return the result table in **any order**.

The result format is in the following example.

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 SELECT E1.name
4 FROM Employee E1
5 JOIN (
6     SELECT managerId, COUNT(*) AS directReports
7     FROM Employee
8     GROUP BY managerId
9     HAVING COUNT(*) >= 5
10 ) E2 ON E1.id = E2.managerId;

```

Saved

Ln 1, Col 1

9. 196. Delete Duplicate Emails

196. Delete Duplicate Emails

Solved

Easy Topics Companies

SQL Schema > Pandas Schema >

Table: Person

Column Name	Type
id	int
email	varchar

id is the primary key (column with unique values) for this table.
Each row of this table contains an email. The emails will not contain uppercase letters.

Write a solution to **delete** all duplicate emails, keeping only one unique email with the smallest `id`.

For SQL users, please note that you are supposed to write a `DELETE` statement and not a `SELECT` one.

For Pandas users, please note that you are supposed to modify `Person` in place.

After running your script, the answer shown is the `Person` table. The driver will first compile and run your piece of code and then show the `Person` table. The final order of the `Person` table **does not matter**.

The result format is in the following example.

MySQL Auto

```

1 # Write your MySQL query statement below
2
3 delete p1 from person p1, person p2
4 where p1.email = p2.email and p1.id > p2.id;

```

Saved

Ln 1, Col 1

10. 1204. Last Person to Fit in the Bus

1204. Last Person to Fit in the Bus

Solved

Medium Topics Companies

SQL Schema > Pandas Schema >

Table: Queue

Column Name	Type
person_id	int
person_name	varchar
weight	int
turn	int

person_id column contains unique values.
This table has the information about all people waiting for a bus.
The person_id and turn columns will contain all numbers from 1 to n, where n is the number of rows in the table.
turn determines the order of which the people will board the bus, where turn=1 denotes the first person to board and turn=n denotes the last person to board.
weight is the weight of the person in kilograms.

There is a queue of people waiting to board a bus. However, the bus has a weight limit of **1000 kilograms**, so there may be some people who cannot board.

Write a solution to find the **person_name** of the **last person** that can fit on the bus without exceeding the weight

MySQL Auto

```
1 # Write your MySQL query statement below
2 SELECT person_name
3 FROM (
4     SELECT person_name, turn,
5           SUM(weight) OVER (ORDER BY turn) AS total_weight
6     FROM Queue
7 ) AS q
8 WHERE total_weight <= 1000
9 ORDER BY turn DESC
10 LIMIT 1;
```

Saved Ln 1, Col 1

11. 176. Second Highest Salary

176. Second Highest Salary

Solved

Medium Topics Companies

SQL Schema > Pandas Schema >

Table: Employee

Column Name	Type
id	int
salary	int

id is the primary key (column with unique values) for this table.
Each row of this table contains information about the salary of an employee.

Write a solution to find the second highest **distinct** salary from the **Employee** table. If there is no second highest salary, return **null** (return **None** in Pandas).

The result format is in the following example.

Example 1:

Input:
Employee table:

MySQL Auto

```
1 # Write your MySQL query statement below
2
3 SELECT (
4     SELECT DISTINCT salary
5     FROM Employee
6     ORDER BY salary DESC
7     LIMIT 1 OFFSET 1
8 ) AS SecondHighestSalary;
```

Saved Ln 1, Col 1